the input AC signal. In practical oscillator circuits, there is loss of some energy in inductor coils and capacitors due to electrical resistance. A small amount of energy is used up in overcoming these losses during every cycle of charging and discharging of the capacitor. Due to this, the amplitude of the oscillations decreases gradually. Hence, the tank circuit produces damped electrical oscillations.

In order to produce undamped oscillations, a positive feedback is provided from output to input by feedback network. This compensates energy loss in tank circuit. The frequency of oscillations is determined by the values of L and C and is given by

$$f = \frac{1}{2\pi\sqrt{LC}} \qquad (10.8)$$

Barkhausen conditions for sustained oscillations

The following conditions called Barkhausen conditions should be satisfied for sustained oscillations in the oscillator.

- There should be positive feedback.
- The loop phase shift must be 0^0 or integral multiples of 2π.
- The loop gain must be unity. That is, $|A\beta| = 1$.

Here, A is the voltage gain of the amplifier, β is the feedback ratio (the fraction of the output that is fed back to the input).

There are different types of oscillator circuits based on the different types of tank circuits. Examples: Hartley oscillator, Colpitts oscillator, Phase shift oscillator and Crystal oscillator.

Applications of oscillators

Transistor oscillators are used

- to generate periodic sinusoidal or non sinusoidal wave forms
- to generate RF carriers
- to generate audio tones
- to generate clock signal in digital circuits
- as sweep circuits in TV sets and CRO

EXAMPLE 10.8

Calculate the range of the variable capacitor that is to be used in a tuned-collector oscillator which has a fixed inductance of 150 μH. The frequency band is from 500 kHz to 1500 kHz.

Solution

Resonant frequency,

$$f = \frac{1}{2\pi\sqrt{LC}}$$

On simplifying, we get

$$C = \frac{1}{4\pi^2 f^2 L}$$

i) When frequency = 500 kHz,

$$C = \frac{1}{4\times 3.14^2 \times (500\times 10^3)^2 \times 150\times 10^{-6}}$$
$$= 676 \text{ pF}$$

ii) When frequency = 1500 kHz,

$$C = \frac{1}{4\times 3.14^2 \times (1500\times 10^3)^2 \times 150\times 10^{-6}}$$
$$= 75 \text{ pF}$$

Therefore, the capacitor range is from 75 to 676 pF.

10.5 DIGITAL ELECTRONICS

Digital Electronics is the branch of electronics which deals with digital signals. It is increasingly used in numerous applications ranging from high end processor circuits to miniature circuits for signal processing, communication etc. Digital signals are preferred over analog signals due to their better performance, accuracy, speed, flexibility and immunity to noise.

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